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Impact wrench having dynamically tuned drive components and method thereof

Abstract

The present invention provides methods and systems an impact wrench having dynamically tuned drive components, such as an anvil/socket combination, and related methodology for dynamically tuning the drive components in view of inertia displacement, as well as stiffness between coupled components, and with regard to impact timing associated with clearance gaps between the component parts.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority of each of the applications recited below and specifically claims the benefit of priority of and is a continuation of U.S. patent application Ser. No. 16/590,296 entitled IMPACT WRENCH HAVING DYNAMICALLY TUNED DRIVE COMPONENTS AND METHOD THEREOF, filed on Oct. 1, 2019, which is a continuation of U.S. patent application Ser. No. 15/290,957 entitled IMPACT WRENCH HAVING DYNAMICALLY TUNED DRIVE COMPONENTS AND METHOD THEREOF, filed on Oct. 11, 2016, which is a continuation-in-part of and claims the benefit of priority from U.S. patent application Ser. No. 13/080,030 entitled ROTARY IMPACT DEVICE and filed on Apr. 5, 2011, and which also claims the benefit of priority and is a continuation-in-part of U.S. patent application Ser. No. 14/169,945 entitled POWER SOCKET FOR AN IMPACT TOOL and filed on Jan. 31, 2014, and which also claims the benefit of priority and is a continuation-in-part of U.S. patent application Ser. No. 14/169,999 entitled ONE-PIECE POWER SOCKET FOR AN IMPACT TOOL and filed on Jan. 31, 2014. U.S. patent application Ser. No. 16/590,296, U.S. patent application Ser. No. 15/290,957, U.S. patent application Ser. No. 13/080,030, U.S. patent application Ser. No. 14/169,945, and U.S. patent application Ser. No. 14/169,999 are incorporated by reference herein in their entireties.

BACKGROUND

Technical Field

(1) The following relates generally to an improved impact wrench, and more generally relates to an improved impact wrench having dynamically tuned drive components, such as an anvil socket combination and corresponding method of optimizing the characteristic functionality thereof.

State of the Art

(2) Impact tools, such as an impact wrench, are well known in the art. An impact wrench is one in which an output shaft or anvil is struck by a rotating mass or hammer. The output shaft is typically coupled to a fastener engaging element, such as a socket, configured to connect with a fastener (e.g. bolt, screw, nut, etc.) to be tightened or loosened, and each strike of the hammer on the anvil applies torque to the fastener. Because of the nature of impact loading of an impact wrench compared to constant loading, such as a drill, an impact wrench can deliver higher torque to the fastener than a constant drive fastener driver.

(3) Ordinarily, a socket is engaged with a polygonally-shaped mating portion of the anvil of an impact wrench, usually a square-shaped portion, and the socket is, in turn, coupled to a polygonally-shaped portion of a fastener, often having mating hex geometry. The socket commonly has a polygonal recess for receiving the polygonal portion of the fastener, thus resulting in a selectively secured mechanical connection. This connection or engagement of the socket to the fastener often affords some looseness allowing for ease of repeated and intended engagement and disengagement of the components because of tolerance clearances or gaps between the components, wherein the gaps can vary in dimension, possibly as a result of manufacturing variation, and affect the timing and/or a spring effect commonly associated with the transfer of energy from the socket to the fastener. Additionally, there is often also a spring effect between the ordinary square-shaped socket and anvil mating connection. Therefore, it is desirable to increase the amount of torque applied by the socket to overcome spring effect, to maximize energy transfer, to increase net effect, and to improve performance of the impact wrench.

SUMMARY

(4) An aspect of the present disclosure includes an impact wrench comprising: a housing, configured to house a motor; a hammer, configured to be driven by the motor; an anvil configured to periodically engage the hammer as it is driven; and a socket having an interface configured to be removably coupled to a corresponding interface of the anvil, wherein the socket is further configured to engage a fastener; and wherein the anvil and socket are tuned and configured so that their combined stiffness, when removably coupled together including the interface between the two, is optimized so as to be between 1.15 and 1.45 times the stiffness of the fastener upon which the impact wrench is being used.

(5) Another aspect of the present disclosure includes an impact wrench comprising: a housing, configured to house a motor and a hammer driven by the motor; an anvil configured to periodically engage the hammer as it is driven; and a socket removably coupled to the anvil, wherein the socket is further configured to engage a fastener; and wherein the anvil and socket are tuned and configured so that their combined inertia, when removably coupled together, is equal to the inertia of the hammer, thereby facilitating a hammer velocity of zero when the socket exerts peak force upon the fastener during tightening.

(6) Still another aspect of the present disclosure includes an impact wrench comprising: a housing; a motor within the housing; a hammer driven by the motor; an anvil configured to engage the hammer; and a socket removably coupled to the anvil, wherein the socket is further configured to engage a fastener; and wherein the anvil and socket are dynamically tuned and configured so that the ratio of the inertia of the combined socket and anvil components and the inertia of the hammer has a specific relationship with the ratio of the anvil/socket combination stiffness and hex stiffness to achieve maximum output at a minimum total weight.

(7) Yet another aspect of the present disclosure includes a method of dynamically tuning the drive components of an impact wrench, the method comprising: modifying the interface between an anvil and a socket so that the combined stiffness of the anvil and socket when coupled together is in the region of $\frac{4}{3}$ the stiffness of the hex fastener on which the impact wrench is being used.

(8) A further aspect of the present disclosure includes a method of dynamically tuning the drive components of an impact wrench, the method comprising: modifying the weight distribution of an anvil and a socket so that their combined inertia, when removably coupled together, is equal to the inertia of a hammer of the impact wrench, thereby facilitating a hammer velocity of zero when the socket exerts peak force upon the fastener during tightening.

(9) Still a further aspect of the present disclosure includes a method of dynamically tuning the drive components of an impact wrench, the method comprising: equating the drive components of the impact wrench with springs and masses in a double oscillator model so that a hex fastener is equated with a first spring force, a socket is equated with a first inertial mass, an anvil is equated with a second spring force, and a hammer is equated with a second inertial mass; and tuning the anvil and socket so that the ratio of the inertia of the combined socket and anvil components and the inertia of the hammer has a specific relationship with the ratio of the anvil/socket combination stiffness and hex stiffness to achieve maximum output at a minimum total weight.

(10) The foregoing and other features, advantages, and construction of the present disclosure will be more readily apparent and fully appreciated from the following more detailed description of the particular embodiments, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members:

- (2) FIG. 1 is a side view of one embodiment of a common impact wrench and standard socket;
- (3) FIG. 2 is a perspective view of the common impact wrench of FIG. 1;
- (4) FIG. 3 is a partial cut-away view of the common impact wrench and standard socket of FIGS. 1 and 2;
- (5) FIG. 4A is a front perspective view of an embodiment of a standard ball and cam anvil mechanism that is often used with a common impact wrench and a standard socket;
- (6) FIG. 4B is a rear perspective view of an embodiment of the standard ball and cam anvil mechanism of FIG. 4A;
- (7) FIG. 5 is a front perspective view of an embodiment of a standard swinging weight or Maurer mechanism that is often used with a common impact wrench and a standard socket;
- (8) FIG. 6 is an exploded perspective view of a drive system of a common impact wrench having a common ball and cam mechanism, wherein the drive components are correlated with and respectively equated into a corresponding double oscillator model;
- (9) FIG. 7 is an exploded perspective view of a drive system of a common impact wrench having a standard swinging weight or Maurer mechanism, wherein the drive components are correlated with and respectively equated into a corresponding double oscillator model;
- (10) FIG. 8 is an exploded perspective view of a drive system of a common impact wrench having a standard rocking dog mechanism, wherein the drive components are correlated with and respectively equated into a corresponding double oscillator model;
- (11) FIG. 9 is a front perspective view of an embodiment of a tuned power socket
- (12) FIG. 10 is a rear perspective view of the embodiment of the tuned power socket of FIG. 9;
- (13) FIG. 11 is a side view of one embodiment of a common impact wrench and tuned power socket;
- (14) FIG. 12 is a partial cut-away view of the common impact wrench and tuned power socket of FIG. 11;
- (15) FIG. 13 is a block diagram modelling fastening operation of a common impact wrench and a tuned power socket having an inertia member that adds a substantial mass a large distance from the axis of rotation of the socket;
- (16) FIG. 14 depicts a plot of energy versus time pertaining to standard non-tuned components of an impact wrench drive system;
- (17) FIG. 15 depicts a plot of energy versus time pertaining to dynamically tuned and optimized components of an impact wrench drive system;
- (18) FIG. 16 depicts a listing stiffnesses of interest and lab-measured ratios;
- (19) FIG. 17A depicts a front perspective view of an embodiment of a dynamically tuned anvil;
- (20) FIG. 17B depicts a rear perspective view of an embodiment of a dynamically tuned anvil;
- (21) FIG. 17C depicts a side view of an embodiment of a dynamically tuned anvil;
- (22) FIG. 18 depicts the mating engagement of dynamically tuned embodiments of an anvil and a socket;
- (23) FIG. 19 depicts a plotted Inertial Ratio vs. Stiffness Ratio curve;
- (24) FIG. 20 depicts the plotted. Inertial Ratio vs. Stiffness Ratio curve of FIG. 19 and includes performance zones pertaining to operable functionality of various tuned and not tuned impact wrench drive systems;
- (25) FIG. 21 depicts a plot of Energy versus Time, when there is no hex clearance between components;
- (26) FIG. 22 depicts a plot of Energy versus Time, when there is hex clearance between components;
- (27) FIG. 23 depicts a plot of Torque versus Time, for a non-stiffened anvil connection;
- (28) FIG. 24 depicts a plot of Torque versus Time for a stiffened anvil connection;
- (29) FIG. 25 depicts a plot of Output Torque vs. Hex Gap comparing a stiffened spline connection and a non-stiffened standard square-shaped connection;

- (30) FIG. 26 depicts a billiard ball model of a larger mass striking a smaller mass;
- (31) FIG. 27 depicts a billiard ball model of a smaller mass striking a larger mass;
- (32) FIG. 28 depicts a billiard ball model of a mass striking another mass having similar inertial properties;
- (33) FIG. 29 depicts a plotted Inertial Ratio vs. Stiffness Ratio curve, along with optimal bounds derived through momentum modelling;
- (34) FIG. 30A depicts a front perspective view of another embodiment of a dynamically tuned anvil;
- (35) FIG. 30B depicts a rear perspective view of another embodiment of a dynamically tuned anvil;
- (36) FIG. 30C depicts a side view of another embodiment of a dynamically tuned anvil;
- (37) FIG. 31 depicts an exploded perspective view of a drive system of a common impact wrench having a tuned ball and cam mechanism, wherein the drive components are correlated with and respectively equated into a corresponding double oscillator model;
- (38) FIG. 32 depicts an exploded perspective view of a drive system of a tuned impact wrench having tuned anvil/socket combination and a standard swinging weight or Maurer mechanism, wherein the drive components are correlated with and respectively equated into a corresponding double oscillator model
- (39) FIG. 33 depicts the structural differences of three cordless impact wrenches having differing tuned components; and
- (40) FIG. 34 depicts various structural features that may be implemented in a stiffened mating engagement of an anvil and a socket.

DETAILED DESCRIPTION OF EMBODIMENTS

(41) Referring now specifically to the drawings, an example of a prior art impact wrench and a common a socket, is illustrated and shown generally in FIG. 1. The socket **1010** may be attached to and driven by an impact tool that is a source of high torque, such as an impact wrench **1012**. The impact wrench **1012** ordinarily includes an output shaft or anvil **1022** having a socket engagement portion **1014** sized for coupling to the socket **1010**. The socket **1010** is intended to be selectively secured to and removably coupled to the impact wrench **1012**.

(42) A common socket **1010** ordinarily has a longitudinal axis **1028** that defines the rotational axis of the socket **1010** when it is secured to the socket engagement portion **1014** of the anvil **1022** of the impact wrench **1012**. The socket **1010** also includes a body **1030** that extends along the axis **1028** from a first longitudinal end **1032** to an opposite second longitudinal end **1034**. An input recess **1038**, which is sized to receive and mate with the socket engagement portion **1014** of the anvil **1022** of the impact wrench **1012**, is defined at the first longitudinal end **1032** of the socket body **1030**. Typically, the recess **1038** is square-shaped to match the standard square-shaped cross-section (see FIG. 2) of the socket engagement portion **1014** of the output shaft or anvil of impact wrench **1012**. It should be appreciated that the square-shaped socket engagement portion **1014** of a common anvil **1022** may have other features, such as, for example, rounded or chamfered edges, or retention features, such as spring loaded balls, O-rings, or other features. In such embodiments, the recess **1038** may be shaped to match the configuration of the socket engagement portion **1014** of the output shaft or anvil **1022** of the impact wrench **1012**.

(43) The socket **1010** normally includes an output recess **1040** that is defined at the opposite second longitudinal end **1034** of the body **1030**. The output recess **1040** is sized to receive a head of a fastener. Typically, the recess **1040** is hexagonal (see FIGS. 3 and 6) to match a common hexagonal-shaped mating portion of a fastener **1**. The fastener **1** may be a nut, screw, bolt, lug nut, etc. It should be appreciated that in other embodiments the output recess **1040** may be configured to receive fasteners having other types of heads, such as, for example, square, octagonal, Phillips, flat, star-shaped or Torx compliant, and so forth. As is well known within the art, at least a portion of the fastener **1** (e.g. a hex-nut, the head of a bolt and the body of a screw) has a polygonal-shape that corresponds with the polygonal-shaped output recess **1040**. During use, the polygonal-shaped

portion of the fastener **1** is inserted into the polygonal-shaped output recess **1040** for operation and is selectively secured to one another, often by friction fit. The socket **1010** is typically made of a durable hard material, such as steel.

(44) As is well known by one of ordinary skill in the art, a typical impact wrench **1012** is designed to receive a standard socket **1010** and designed to deliver high torque output with the exertion of a minimal amount of force by the user. As shown in FIGS. **1-3**, a common impact wrench **1012** normally includes a housing **1016** that encases a motor **1018**. The motor **1018** is often configured to be driven by a source of compressed air (not shown), but other sources of power may be used. Those sources may include electricity, hydraulics, etc. In operation, the motor **1018** accelerates a mass such as, for example, a hammer **1020** that is configured to spin and generate rotating inertia storing energy. This rotating inertia spends a period of time accelerating freely until periodically a clutch of substantial material suddenly interrupts and kinetically locks the rotating mass to the bolt or nut through the anvil **1022** and a socket **1010** connected in series with the anvil **1022**. The high torque output is, therefore, accomplished by storing kinetic energy in a rotating mass, such as a hammer **1020**, and then delivering the energy to a fastener engaged with a socket **101**, which is in turn engaged with an output shaft or anvil **1022** of the impact wrench **1012**. The hammer **1020** is configured to suddenly strike, contact, or otherwise engage the output shaft or anvil **1022**. The sudden engagement of the hammer **1020** with the anvil **1022** creates a high-torque impact. In the illustrative embodiment, the hammer **1020** is configured to slide within the housing **1016** toward the anvil **1022** when rotated. A spring (not shown) or other biasing element may bias the hammer **1020** out of engagement with the anvil **1022**. Once the hammer **1020** impacts the anvil **1022**, the hammer **1020** of the impact wrench **1012** is designed to freely spin again. As shown in FIGS. **1-3**, the impact wrench **1012** also includes a trigger **1024** that is moveably coupled relative to the housing **1016**. In use, compressed air, electric power, or hydraulic fluid, etc. is delivered to the impact wrench **1012** when the trigger **1024** is depressed.

(45) Those of ordinary skill in the art appreciate that there are many known hammer **1020** designs, and also recognize that it is important that the hammer **1020** is configured to spin relatively freely, impact the anvil **1022**, and then spin relatively freely again after impact. In some common impact wrench **1012** designs, the hammer **1020** drives the anvil **1022** once per revolution. However, there are other impact wrench **1012** designs where the hammer **1020** drives the anvil **1022** twice per revolution. The partial cut-away view of the impact wrench **1012** depicted in FIG. **3** reveals a standard ball and cam mechanism hammer and anvil design. FIGS. **4A** and **4B** respectively depict front and rear perspective views of a standard ball and cam anvil **1022**. An embodiment of the common square socket mating engagement portion **1014** is prominently shown in FIG. **4A**, while both FIG. **4A** and FIG. **4B** show how the anvil jaws **1087** extend radially from a central axis of the anvil **1022**. The portion of the anvil **1022** that extends between the jaws **1087** and the square-shaped socket engagement portion **1014** functions as a bearing journal and helps align and support the anvil **1022** during use. A ball and cam anvil **1022** is often utilized in an impact wrench powered by an electric motor. Another common anvil embodiment is shown in FIG. **5**, which depicts a standard swinging weight or Maurer mechanism anvil **3022**. This common type of anvil **3022** includes the typical square-shaped socket engagement portion **3014**. Near the other end of the anvil **3022** are jaws **3087** that are relatively small in diameter, as compared to the common ball and cam anvil **1022** of FIGS. **1-4B**. A common Maurer mechanism anvil **3022** is typically utilized in conjunction with a pneumatically-powered impact wrench. In addition, a Maurer mechanism like anvil **3022** may permit operation with a double hammer design.

(46) The output torque of an impact wrench, such as impact wrench **1012**, can be difficult to measure, since the impact by the hammer **1020** on the anvil **1022** is a short impact force. In other words, the impact wrench **1012** delivers a fixed amount of energy with each impact by the hammer **1020**, rather than a fixed torque. Therefore, the actual output torque of the impact wrench **1012** changes depending upon the operation. An anvil, such as anvil **1022** or **3022** is designed to be

selectively secured to a socket, such as socket **1010**. This engagement or connection of the anvil, such as anvil **1022**, **3022**, to the socket, such as socket **1010**, results in a spring effect when in operation. This spring effect stores energy and releases energy. Additionally, there is a spring effect between the socket **1010** and the fastener **1** to which it is engaged. Again, this spring effect stores energy and releases energy.

(47) It may be beneficial to model the spring effects associated with tightening fasteners using an impact wrench. As is known to one of ordinary skill in the art, the combination of two masses ($m_{sub.1}$ and $m_{sub.2}$) and two springs ($k_{sub.1}$ and $k_{sub.2}$) is often referred to as a double oscillator mechanical system. In this system, the springs ($k_{sub.1}$ and $k_{sub.2}$) are designed to store and transmit potential energy. The masses ($m_{sub.1}$ and $m_{sub.2}$) are used to store and transmit kinetic energy. The drive system or drive components and mechanisms of common impact wrenches can typically be broken down into common fundamental elements. Ordinarily, the drive system is composed of a motor, a hammer, an anvil, a socket, and a joint (or fastener component that is to be driven). The motor can be directly or indirectly coupled to a hammer. The hammer often engages an anvil having mating jaws spaced apart from the center of rotation. The anvil is coupled to a socket with a mating geometric shape, usually a square, and the socket is usually coupled to the nut of the joint with mating hex geometry. As depicted in FIGS. **6-9**, three common impact wrench drive mechanisms are shown in exploded perspective view with drive components respectively modelled. For example, FIG. **6** depicts an exploded perspective view of a drive system of an impact wrench having a common ball and cam mechanism, with components similar to those depicted in FIGS. **1-4B**, wherein the drive components are correlated with and equated into a double oscillator model. As modelled, the joint or hex fastener **1** is equated with a first spring $k_{sub.1}$. The standard socket **1010** is equated with a first inertial mass $m_{sub.1}$. The common ball and cam anvil **1022** is equated with a second spring $k_{sub.2}$, and the associated ball and cam hammer **1020** is equated with a second inertial mass $m_{sub.2}$. The arrows in FIGS. **6-8** are provided for purposes of clarity, primarily to show how each respective mechanical component has a corollary model component.

(48) A common impact wrench drive system employing a standard swinging weight or Maurer mechanism is particularly depicted and modelled in FIG. **7**. The socket **1010** and hex fastener **1** may be configured the same as or similar to those depicted in FIG. **6**, but the swinging weight Maurer mechanism differs, inter alia, from the standard ball and cam mechanism, in that it employs a dual hammer component **3020** and a generally cylindrical anvil **3022** having jaw features correspondingly configured to engage the dual hammers **3020**. The dashed-line box is provided for purposes of clarity, to surround and, thereby, designate the component features of the hammer **3020**. Another well-known impact wrench drive system employing a standard rocking dog mechanism is particularly depicted in FIG. **8**. Again, the socket **1010** and hex fastener **1** may be configured the same as or similar to those depicted in FIGS. **6** and **7**. In a similar manner, a dashed line is provided to delineate the components of the rocking dog hammer **4020**. The anvil **4022** is also generally cylindrical with jaw features configured to engage the rocking dog hammer **4020**. The socket mating end of the anvil **4022** is a standard square shape, and, in a similar manner, the socket mating ends of the anvils **1022** and **3022** depicted respectfully in FIGS. **6** and **7** are also provided with a standard square shape.

(49) For purposes of modelling, the common square drive anvil inertia is extremely low relative to the other components and is treated purely as a torsional spring. The compliance of the drive connection between the socket and the anvil is lumped into the total stiffness of the rest of the anvil and, for purposes of further modelling, will be assumed to be included in the term “anvil stiffness” and will be discussed later. The socket, such as socket **1010** is of relatively high stiffness but relatively large in inertia and is therefore treated as a pure inertia. For the sake of mathematical modeling the joint (or hex fastener **1**) is assumed to be in the “locked” condition, i.e. unable to be moved further, allowing the hex interface to be modeled as a very stiff spring. It is the point at

which the tool cannot move the hex any further that will characterize the “power” of the system. This is true in practice as well. A weak tool normally reaches a locked hex in a relatively short angle and the installed torque is low, whereas a strong tool normally reaches locked hex in a larger angle and achieves a higher installed torque on the same bolt.

(50) The double oscillator system can be tuned to efficiently and effectively transfer energy from the impact device or hammer (modelled as $m_{sub.2}$) through the anvil-socket connection (modelled as $k_{sub.2}$), the socket (modelled as $m_{sub.1}$) and socket-fastener connection (modelled as $k_{sub.1}$) and into the joint fastener **1**. Proper tuning can help ensure most of the energy delivered by the impact wrench hammer $m_{sub.2}$ is transferred through the anvil-socket connection spring $k_{sub.2}$ and into the socket $m_{sub.1}$. During use, the rate of deceleration of the inertial mass of the socket $m_{sub.1}$ is very high since spring $k_{sub.1}$ is stiff. Since deceleration is high the torque exerted on the fastener is high.

(51) One way to tune the drive components of an impact wrench is to increase the inertial mass of the socket; to create a power socket. This can be done, inter alia, by providing the socket with an inertial feature, such as for example an annular ring located a radial distance away from the central axis of the socket. As depicted in FIGS. **9** and **10**, the annular ring may act as an inertial member **2036** increasing the inertial mass of socket **2010**. The purpose of the inertia member **2036** is to increase the overall performance of an impact wrench, by increasing the net effect of the rotary hammer inside the impact wrench, such as rotary hammer **2020** of impact wrench **2012** depicted in FIGS. **11** and **12**. The impact wrench **2012** may be similar to an impact wrench **1012**, and may include similar component elements, such as a housing **2016**, a motor **2018**, a trigger **2024**, and an anvil **2022** having a standard square-shaped socket engagement portion **2014**. The socket **2010** may include a square shaped input recess **2038**, which is sized to receive and mate with the standard square-shaped socket engagement portion **2014** of the anvil **2022** of the impact wrench **2012**. As depicted, the drive mechanism is a common ball and cam mechanism, but any drive mechanism having a square-shaped socket engagement component may be operable with and tunable for improved performance through use of a power inertia socket, such as socket **2010**. The socket **2010** may also include an output recess **2040** that is sized to receive a head (typically a hex head) of a fastener **1**. The performance is increased as a result of the inertia member **2036** functioning as a type of stationary flywheel on the socket **2010**. Stationary flywheel means the flywheel is stationary relative to the socket **2010**, but moves relative to the anvil **2022** and the fastener **1**. By acting as a stationary flywheel, the inertia member **2036** increases the amount of torque applied to the fastener **1** for loosening or tightening the fastener.

(52) In reference to the disclosed tuned power socket embodiments, as illustrated in FIG. **13**, the inertia member **2036** adds a substantial mass a large distance from the axis of rotation **2028** of the socket **2010**. It should be noted that FIG. **13** is shown and modelled in a linear mode, but the impact wrench and socket is a rotary system. Nevertheless, the socket **2036** having inertia member **2036** is represented by $m_{sub.1}$. The socket having inertial member $m_{sub.1}$ is operationally situated between spring effects $k_{sub.1}$ and $k_{sub.2}$; in other words, the socket connects with both the fastener **1** (modelled as spring effect k_1) and the anvil (modelled as spring effect $k_{sub.2}$). Hence, the spring rate of the common square-shaped anvil and socket connection is represented by $k_{sub.2}$ and the spring rate of the socket and fastener connection is represented by while the fastener itself is represented by ground. The mass moment of inertia of the impact wrench is designated $m_{sub.2}$ and represents the mass moment of inertia of the rotary hammer inside the impact wrench. With respect to the tuned power socket **2010**, the spring rate of $k_{sub.1}$ is three times that of $k_{sub.1}$ and $k_{sub.2}$ combined, causing very high torques to be transmitted from the socket **2010** having an inertia member (modelled as $m_{sub.1}$) to the fastener.

(53) When impact wrench drive system tuning is focused primarily on the socket, the tuning process operates under the notion that there is an optimal socket inertia for a given combination of mechanism inertia and joint and anvil stiffness. As such, the elements of the double oscillator

system are predetermined. The rotary hammer inside the impact wrench $m_{sub.2}$ and springs $k_{sub.1}$ and $k_{sub.2}$ are assumed to have defined values. For tuning the system with the focus primarily on the socket, the only value which needs to be determined is the inertia member $m_{sub.1}$ **2036** of the socket **2010**, in order to achieve socket-optimized inertia. A common impact wrench, depending upon the drive size (i.e. $\frac{1}{2}$ ", $\frac{3}{4}$ ", 1"), has a different optimal inertia for each drive size. The spring rate $k_{sub.2}$ and the rotary hammer inertia m ; inside the impact wrench are substantially the same for all competitive tools of similar drive size incorporating common drive mechanisms, such as, for example, those impact wrench drive systems depicted and modelled in FIGS. 6-8. However, while a tuned power socket significantly increases impact wrench drive system performance, the non-tuned components, such as the hammer, the anvil, and the socket still may dynamically harbor unused energy, thereby preventing full power transfer from the impact wrench to the fastener joint being worked upon. The tuning of the impact wrench drive system is not fully realized or optimized.

(54) To fully, and even optimally, tune an impact wrench drive system, focus may also be placed on the anvil and socket combination, two important components of an impact wrench drive system, and tuning methodology may consider optimizing the characteristics of each of the impact wrench drive system components that function together, to not only to have a stronger interconnection between the parts but to also perform at a higher level without introducing additional power input. Such optimal impact wrench tuning methodology introduces the concept of dynamic manipulation of both the socket inertia and the anvil-socket stiffness, in order to minimize the socket inertia for maximum output, thereby minimizing total tool weight and size. Dynamic impact wrench tuning, therefore, contemplates the ratio of the inertia of the combined socket and anvil components, as well as the inertia of the impacting mechanism, and considers how drive system performance has a specific relationship with the ratio of the anvil/socket combination stiffness and hex stiffness to achieve maximum output at the minimum total weight. The theory behind tuning the power socket, and in particular the methodology associated with determining the optimal component inertia of the socket still applies. The difference is the introduction of an additional independent variable.

(55) When dynamically tuning impact wrench drive components, focus may be placed on the behavior of the various drive system elements, when in contact as a result of the collision of the hammer with the anvil from the moment of first contact until the moment the energy has reached and transferred to the bolt hex. At the beginning of this energy transfer period the hammer inertia has some initial velocity which represents all the kinetic energy that any particular impact can possibly have. When initial contact between the anvil and hammer jaws occurs, there is ordinarily a measurable amount of rotational clearance between the engaged components that must be consumed before any energy can transfer. There may be rotational play between anvil and socket, particularly if that connection is facilitated by the common square-shaped geometry. There is also rotational play between the internal hex of the socket and the external hex of the nut. Depending on how one chooses to consider the rotational clearance typically existent between impact wrench drive system components, there are two primary tuning models that may be implemented to fully, and even optimally, tune the impact wrench drive system. The optimal cases for each of the tuning models serve to provide upper and lower bounds for dynamically tuned impact wrench drive system performance.

(56) Spring-Mass Oscillator Model

(57) For the purposes of this model and related discussion, an assumption is made that rotational play or clearance gaps between impact wrench drive system components has no significant effect on the behavior of the drive system and is assumed to be completely consumed. As the impact wrench drive system components wind up, all of the mechanical elements having varying amounts of inertia and stiffness contribute to a relatively complex oscillatory behavior. Energy is transferred from each spinning inertia to each series spring element, and kinetic energy converts to potential energy and back again in what may seem somewhat chaotic in the span of milliseconds. Tuning

methodology focused primarily on modifying a socket to create a tuned power socket has taught us that choosing the inertia of the socket to be substantially higher than that of currently available standard sockets can enhance the transfer and concentration of the energy into the joint without increasing the energy put into the system. Understanding the relationships between these parts and the effects of their inertias and associated stiffness when interacting with each other, and the delivery of energy through the system, is critical to dynamically optimizing the impact wrench system to deliver as much energy to the fastener joint as possible.

(58) The dynamic tuning and optimization process for socket and anvil inertia and stiffness of the various component connections of the impact wrench/fastener joint system begins with a calculation of the system modeled as lumped masses and springs where there is no rotational play or clearance gaps in between the components and the components are connected rigidly when they first come into contact. For the energy transfer time period in question, this assumption is reasonable and helpful to simplify the motion formulas. A typical schematic diagram for modelling a standard air driven impact wrench is shown in the diagrams depicted in FIGS. 6-8. While the anvil configurations, clutch mechanisms and actual equations of motion differ slightly between the various drive mechanisms, the theory and modelling methodology is largely the same.

(59) As discussed previously, a typical square-shaped anvil/socket mating connection has relatively low inertia, and the compliance of the anvil/socket connection is lumped into a total “anvil stiffness.” In an ideal case where the designer has complete control over all elements of the system, including the hex, there is a closed form solution to the positions, velocities and accelerations of the spring-mass oscillators shown in FIGS. 6-8. It is as follows, where “x” is the rotational angle, w refers to the angular velocity, f refers to an initial angle and “a” and “C” are constants associated with amplitude. The subscripts 1 and 2 refer to the socket and hammer inertial bodies respectively. Modelling equations may be set forth as follows:

$$x_{sub.2} = C_{sub.1} a_{sub.21} \sin(\omega_{sub.1} t + \phi_{sub.1}) + C_{sub.2} a_{sub.22} \sin(\omega_{sub.2} t + \phi_{sub.2})$$

$$x_{sub.1} = C_{sub.1} a_{sub.11} \sin(\omega_{sub.1} t + \phi_{sub.1}) + C_{sub.2} a_{sub.12} \sin(\omega_{sub.2} t + \phi_{sub.2})$$

Equations 1

(60) The initial conditions of the impact wrench drive system are given as: $x_{sub.1} = x_{sub.2} = 0$ The arbitrary origins of angular position of the inertias is zero $v_{sub.1} = 0$ The anvil and socket start each impact stationary $v_{sub.2} < > 0$ This is the angular velocity of the hammer after being accelerated by the motor and is treated as a known constant

(61) For this set of initial conditions, the constants “a” and “C” are as follows:

$$(62) \quad C_1 = \frac{-a_{12} v_{20}}{(a_{22} a_{11} - a_{12} a_{21})_1} \quad \text{Equation 2} \quad C_2 = \frac{-a_{11} v_{20}}{(a_{11} a_{12} - a_{22} a_{11})_2}$$

(63) The phase angles Φ are zero and the a's describe modal shapes:

$$(64) \quad a_{21} = \frac{k_2 a_{11}}{k_2 - m_2^2} \quad \text{Equation 3} \quad a_{22} = \frac{k_2 a_{12}}{k_2 - m_2^2}$$

(65) The following assignment may be made:

$$a_{sub.11} = a_{sub.12} = 1$$

(66) Which, then reduces the “C” and “a” constants to:

$$(67) \quad C_1 = \frac{v_{20}}{(a_{21} - a_{22})_1} \quad \text{Equation 4} \quad C_2 = \frac{-v_{20}}{(a_{21} - a_{22})_2} \quad a_{21} = \frac{k_2}{k_2 - m_2^2} \quad \text{Equation 5} \quad a_{22} = \frac{k_2}{k_2 - m_2^2}$$

(68) Where the natural frequencies $\omega_{sub.1}$ and $\omega_{sub.2}$ are given by:

$$(69) \quad \omega_{1,2} = \frac{1}{2} \left\{ \frac{k_1}{m_1} + \frac{k_2}{m_2} \left(1 + \frac{m_2}{m_1} \right) - / + \sqrt{\left[\frac{k_1}{m_1} + \frac{k_2}{m_2} \left(1 + \frac{m_2}{m_1} \right) \right]^2 - \frac{4k_1 k_2}{m_1 m_2}} \right\} \quad \text{Equation 6}$$

(70) Equations 1 to 6 describe the motion of the mass under some initial conditions and any set of spring constants and inertias. In the ideal case where the designer has control of all inertias and stiffnesses, the specific values of those quantities can be determined by applying some dynamic energy accounting conditions throughout the impact cycle. Maximum deflection of spring $k_{sub.1}$, or the peak energy in the hex, would occur when all other components had completely given up their energy at the precise time when $k_{sub.1}$ reached its peak energy. This means that the hammer

and socket have no kinetic energy and hence zero velocity and the anvil, spring $k_{sub.2}$, has no potential energy and hence, no deflection. Again, this is the ideal case.

(71) To find the optimal torque in spring $k_{sub.1}$, the following conditions may be applied to Equations 1-6:

(72) At some subsequent time $t=A$.

(73) $v_{sub.1A}=v_{sub.2A}=0$ When the nut hex ($k_{sub.1}$) is at its peak torque, the velocity of the hammer and the socket are zero. Otherwise, there would be energy tied up in those components. $x_{sub.2A}-x_{sub.1A}=0$ The anvil deflection must also be zero, otherwise there will be energy tied up in the anvil that should be in the hex. $x_{sub.1A} < > 0$ The hex “spring” is deflected.

(74) For this set of conditions to be true:

(75) $m_1 = \frac{3}{4}m_2$ $k_1 = \frac{3}{4}k_2$

(76) The above results describe the optimal inertia and the optimal stiffness of the components that reside between the hammers and the hex under ideal conditions. Hence, when dynamically tuning an impact wrench drive system, in a perfect world, the inertia of the socket/anvil combination must be $\frac{3}{4}$ of the combined inertia of the hammer components of an impact mechanism at the same time the stiffness of the nut hex must be $\frac{3}{4}$ of the stiffness of the anvil for maximum output and minimum total weight.

(77) It may be helpful to visually demonstrate the difference between a standard impact wrench drive system having a square-shaped anvil/socket connection, such as the impact wrench drive systems depicted and modelled in FIGS. 6-8, and a dynamically tuned and optimized ideal system. The demonstration charts the energy that is contained in each element of the impact wrench drive system at any time during the energy transfer period. As depicted in FIG. 14, the performance of the standard impact wrench system is plotted for Energy versus Time. From the chart, it is apparent that when the hex spring ($k_{sub.1}$ —star-marked dashed line) is at its peak, there is a noticeable amount of energy that still remains in the other components. At the time the hex (k_1) reaches its peak, the anvil (k_2 —square-marked dashed line) still contains a significant amount of energy. This non-transferred energy is a source of inefficiency that can be remedied by the dynamic tuning of the inertia and stiffness elements in the impact wrench drive system.

(78) With regard to the dynamically tuned and optimized impact wrench drive system, FIG. 15 graphically depicts the performance of the tuned wrench via an Energy versus Time plot. In the optimized drive system shown in FIG. 15, the hammer, anvil and socket have all released their energy at the precise time that the hex (star-marked dashed line) reaches its peak. This notable that the peak is more than 100% greater than the standard tool output. This dynamically tuned impact wrench drive system is indeed “ideal,” as all the incoming hammer energy makes its way into the hex and then 100% returns to the hammer by $t=3.5E-04$ (circle-marked solid line). Those of ordinary skill in the art will appreciate that, in reality, some energy will be lost in both cases to friction. However, the dynamic tuning of the ideal case, in some ways, sets a bound on performance that provides input for practical real life tuning.

(79) While the inertia of the socket can readily be increased or decreased through introduction of part-geometry changes, such changes may result in unwanted adverse effects on the overall weight of the tool and the ability to access tight spaces where bolts or other fasteners might be located. Achieving the optimum stiffness is much more challenging than the ideal case for at least two reasons: 1) there are many nut sizes in existence on which the impact wrench will likely be used, which presents the possibility for a wide range of stiffness ratios to a given tool—a decision must, therefore, be made regarding a hex size for which to optimize; and 2) the anvil stiffness, which includes the stiffness of the interface between the anvil and the socket can be quite low, as in the currently available common square-shaped interface impact wrenches, such as those with drive systems depicted and modelled in FIGS. 6-8, in comparison to the hex stiffness available in an ideal case. Lab experimentation has helped to clarify real world causality associated with modifications made to the anvil/socket inertia and stiffness ratios. For example, FIG. 16 depicts a

Table 1 listing stiffnesses of interest and lab-measured ratios. Since the common square-shaped interface resides in series with the anvil body, the overall stiffness will always be lower than the lowest stiffness in the series, due to the reciprocal rule, as set forth mathematically below:

$$1/K_{\text{sub.Total}} = 1/K_{\text{sub.square}} + 1/K_{\text{sub.anvil}}$$

$$K_{\text{sub.Total}} = 1 / (1/K_{\text{sub.square}} + 1/K_{\text{sub.anvil}}) \quad \text{Equations 7}$$

(80) So, when the lab-measured data for a common Maurer mechanism impact wrench with a standard square-shaped interface (See FIG. 16) is applied to the equation, we get the following:

$$K_{\text{sub.Total}} = 1 / (1/274,000 + 1/55,000) = 46,000$$

(81) In order to achieve an optimal stiffness ratio, the total anvil stiffness (including the interface with the socket) needs to be $4/3 * K_{\text{sub.1}}$. With regard to a 15/16" hex fastener, as set forth in the data listed in Table 1 of FIG. 16, $K_{\text{sub.1}} = 335,000$ in-lb/rad. Hence $4/3 * K_{\text{sub.1}}$ renders an optimal $K_{\text{sub.Total}}$ of approximately 446,700 in-lb/rad. Since the square-shaped interface itself is much less than that, it is impossible to achieve the required stiffness even if the stiffness of the body of the anvil was increased 10 fold. Thus, the most effective way to achieve the required stiffness is to increase the interface stiffness well above the requirement for the total, so that the addition of the body of the anvil brings the total down to the optimal number. Lab testing has confirmed that a stiff interface, such as a spline interface, accomplishes this. Through a Finite Element Analysis, a 24 tooth 20/40 pitch spline has been determined to have a measured stiffness of approximately 1,800,000 in-lb/rad. Hence, using Equation 7 and solving for the anvil stiffness, the stiffness of the anvil body can be determined as follows:

$$1/K_{\text{sub.anvil}} = 1/K_{\text{sub.Total}} - 1/K_{\text{sub.spline}}$$

$$K_{\text{sub.anvil}} = 1 / (1/K_{\text{sub.Total}} - 1/K_{\text{sub.spline}})$$

$$K_{\text{sub.anvil}} = 1 / (1/446,700 - 1/1800K)$$

$$K_{\text{sub.anvil}} = \text{approx. } 594,000 \text{ in-lb/rad}$$

(82) As determined, this stiffness is a significant increase over the standard anvil. However, a common cordless impact mechanism, often referred to as a "Ball and Cam" type mechanism, lends itself well to the geometry changes required to meet this requirement. The jaws of the corresponding hammer are spaced relatively far apart, which allows the anvil diameter to increase to not only better support the jaws, but a larger anvil diameter also increases the associated anvil inertia, thereby tuning the device and meeting the optimal inertia requirement. One such tuned anvil 5022 embodiment is depicted in various perspective views in FIGS. 17A-C. As embodied, the tuned anvil 5022 includes an internal splined socket mating recess 5047 (See particularly FIG. 17A), which is configured to receive an externally splined portion of a tuned socket. While this embodiment specifically includes 24 spline teeth having a 20/40 pitch, those of ordinary skill in the art will appreciate that the tuned socket may have varying numbers of spline teeth with differing pitches. The diameter D1 surrounding the splined socket mating recess 5047 may function as a bearing journal 5085, at a much larger diameter than a standard anvil 1022 (see FIGS. 1-4B), and also as a portion of the m.sub.1 inertia for the tuning process. The jaws 5087 may be structurally and functionally similar to jaws 1087 of a standard ball and cam anvil 1022. The neckdown area 5089, clearly depicted in FIG. 17C, serves to control the stiffness of the tuned anvil 5022. The diameter and length of the neck play a significant role in the stiffness of the tuned anvil 5022. The hole visible in FIG. 17B serves as a bearing journal to support other components of an impact wrench, in a manner similar to the functionality of a similar bearing journal in the standard ball and cam anvil 1022.

(83) A tuned anvil 5022 is depicted, in FIG. 18, as engaged with a correspondingly tuned and configured socket 5020. The socket 5020 may have an externally splined portion 5017 configured to mate with the internally splined socket mating recess 5047 of a tuned anvil 5022. The colors (or gradient shading) depicted in FIG. 18 represent the deflection data collected using a Finite Element Analysis program. The inertial sum of the socket 5020 inertia and the large diameter end of the anvil 5022 in front of the neck 5089 are determined using the optimization process described herein

above. The inertia of the anvil can be increased by lengthening the spline portion of the anvil (L1 of FIG. 17C) and/or increasing the diameter (or thickness) surrounding the spline (Da, again of FIG. 17/c). In general, the greater the anvil inertia can be, the smaller and less bulky the sockets need to be. The trade-off of socket and anvil size with the increase in output needs to be evaluated on a case by case basis.

(84) Dynamic impact wrench drive system tuning involves a determination, based on mathematical modelling as assisted by empirical data, of optimum trade-offs between inertia and stiffness. As depicted in FIG. 19, a plotted output is generated by numerically solving applicable differential equations at various inertias and stiffness levels using an iterative optimization algorithm based on knowledge gained from empirical data. The x-axis is the designed ratio of the anvil stiffness to the hex stiffness. The y-axis is for the Inertia ratio of anvil-socket combination to mechanism. When any three quantities are known, the fourth quantity can be determined by finding the intersection with the curve. Anywhere that is NOT on the curve has lower tool output and potentially more weight than it could otherwise have. For example, at a stiffness ratio of 0.5, the required Inertia ratio from the curve would be about 1.2. Multiply the mechanism inertia by 1.2 and that is the target inertia for the m.sub.1 body or, in the case of a spring oscillation model based tuning process, the optimal socket/anvil combination. Any inertia level above or below that level would result in lower output of the system. There are several things worth noting about the plotted inertial Ratio vs. Stiffness Ratio curve. In the region below 0.5 stiffness ratio, the curve is very steep and requires a large amount of added inertia to optimize. As the stiffness ratio increases, either by increasing anvil stiffness or decreasing the hex stiffness/size, the required inertia ratio for optimization drops significantly. Above the stiffness ratio of 1, the curve is much flatter and requires a relatively low range of inertia to be optimal. With respect to tuning an anvil, a hex stiffness ratio of precisely 1.33 (or a reciprocal hex-anvil ratio of 0.75) results in an optimal inertia ratio (anvil-socket to mechanism) of 0.75 and corresponds to the local minimum on the plotted curve. This numerical solution agrees with the closed-form solution where both are dynamically optimized simultaneously where:

$$(85) m_1 = \frac{3}{4}m_2 \quad k_1 = \frac{3}{4}k_2$$

(86) The Inertia Ratio vs. Stiffness Ratio plot can be very insightful for tuning purposes, especially when utilized in conjunction with empirical data pertaining to impact wrench drive systems. For example, as depicted in FIG. 20, the vertically cross-hatched area in the same plot is where most common anvils and standard square-shaped sockets operate today. These offerings are nowhere near the optimal curve because the ordinary anvils are low in stiffness and typical sockets are very low in inertia relative to their mechanisms. A tuned power socket for standard square drive tools operates in the diagonally cross-hatched region. The anvils are the same (meaning the anvils are not optimized and include common square-shaped socket mating portions), but the inertia has been increased significantly and, for a narrow range of hexes, the inertia is perfectly tuned. The region that is lightly shaded is where dynamically tuned drive systems will likely operate most frequently, particularly with certain mechanism types, where the stiffness of the anvil can be increased to a point where the optimal inertia ratio is between 0.75 and 1.0 making the final tool power to weight ratio extremely hard to compete with. Where the anvil cannot be designed to be significantly higher in stiffness, the region designated by horizontal cross-hatching will likely be the target area for optimization.

(87) When tuning impact wrench drive system performance through employment of spring-mass oscillation modelling, the Inertia Ratio vs. Stiffness Ratio plot can be used to determine the optimal inertia for ANY stiffness ratio that is achieved. There are performance advantages associated with moving the stiffness ratio as close to 1.33 as possible, so that the inertia can be as low as possible and still perform at the highest level. The stiffness of the interface between the socket and the anvil will determine the extent to which the required inertia can be split between the socket and the anvil. In the case of a square drive connection, both the model and empirical data have demonstrated that

the connection is not stiff enough to treat the anvil and socket as a single mass and, therefore, the substantial part of the required inertia may be contained in the socket, such as in the design of the tuned Power Socket. With a stiffer connection, such as with the implementation of a spline drive connection, the inertia may be divided in any convenient manner between the two components thereby reducing the potential for reduced access due to the added material in the socket.

(88) As set forth above, the ideal case is interesting, but it is the exception and not the rule. There are various reasons why the impact dynamics do not operate with all parameters at their optimum. Firstly, a user is likely to use an impact wrench on a wide range of hex sizes. Each hex size will exhibit a wide variety of stiffness behaviors. Large hexes will appear very stiff while small hexes will appear relatively soft. Estimating what stiffness to expect for a given hex size is a matter of experimentation and empirical testing. In the end, however, there will likely end up being hex sizes for which the tool is not fully optimized. As depicted in FIGS. 19 and 20, the shape of the optimality curve, however, allows the designer to optimize at a relatively large hex size resulting in a very close to optimal condition for hex sizes below the hex for which the tool was optimized. Secondly, there are physical limitations to the anvil for each mechanism that may preclude the achievement of high levels of stiffness. For example, long and thin rods can typically be made stiffer by becoming shorter and/or larger in diameter. In the case of the anvil depicted in FIG. 5, which is very often used in air driven impact wrenches, the stiffness is limited by the portion of the anvil with the jaws that has a relatively small diameter and is much longer than it is in diameter. Thirdly, there are losses in the system that will always remove energy and cause the idealized equations to over/underestimate the parameters. The model can be refined to include these elements over time to enhance its prediction capability. For anvils like the standard Maurer mechanism depicted in FIG. 5, the optimum stiffness may not likely be achieved for many of the of hex sizes on which it was designed to operate. For sub-optimal stiffness, the differential equations mentioned earlier may be solved numerically omitting the assumptions that the potential and kinetic energy contained in the hammer, anvil and socket are all zero when the hex energy is at its peak. It becomes an iterative and dynamic optimization process that may take into consideration the known quantities of hammer inertia, initial hammer velocity, designed anvil stiffness and prescribed hex stiffness to drive the unknown quantity of socket inertia to maximize the $k_{sub.1}$ torque.

(89) The spring-mass oscillation model includes assumptions requiring the hammer, anvil, socket and hex nut fastener to be in contact for the duration of the impact event. Even if the forces arise during the simulation, the math of the model does not contemplate the separation of the component elements. Test data has revealed that this is actually a relatively rare case in actual practice, but certainly a possible and potentially bounding case.

(90) Momentum Model

(91) There are several other possible combinations of state of contact that can affect the optimization of the inertia and stiffness parameters of an impact wrench drive system under a designer's control. The ability of the components to disengage from each other is commonly afforded by the loose fits between the anvil and the socket as well as between the socket and the hex nut. The loose fits are, to some extent, required in order to allow for manufacturing variation and ease of repeated and intended assembly and disassembly during normal use. These loose clearances or gaps between the impact wrench drive system components are critical to consider with regard to the energy transfer abilities of the drive system due to their more common presence at the time when hammer contact with the anvil is initiated.

(92) The clearances between the impact wrench drive system components have an important role in the timing of the energy transfers that occur between the parts. In this momentum model analysis, it is helpful to think of the impact event, not as an instantaneous or discontinuous change in state, but rather a motion defined by rapidly changing accelerations that depend on what is in contact at that time. In the spring-mass oscillation model, the energy that each component contains during the impact event was described. In a zero-clearance scenario, as the spring-mass oscillation model

described, there can be energy harbored in the various components at the time when it is desired for all of the energy to arrive at the interface between the socket and the nut. The harbored energy arrives late (if at all) and is unable to do any valuable work on the nut. In a clearance gap, or momentum model, scenario where there is an angle through which the socket (or anvil) is required to cross before any contact occurs, there can be substantially more time for the energy transfer between the bodies to occur BEFORE the nut/socket interface reaches its peak torque. A depiction of how the energy contained in the hammer, anvil, socket and hex at any given time for a given set of design and initial condition parameters is depicted in FIGS. 21 and 22. In particular, FIG. 21 depicts a representation with all components in contact and unable to disengage, while FIG. 22 introduces a large gap between the socket and the nut and allows all components to separate when the forces allow it to occur. It is notable that, as plotted in FIG. 21, when the star-marked solid line associated with $k_{sub.1}$ reaches its peak, other components still contain energy. Yet, as plotted in FIG. 22, when the star-marked solid line associated with $k_{sub.1}$ peaks, all other levels are zero. Note also that the peak of the star-marked solid line associated with $k_{sub.1}$ is higher and occurs later in time than the peak in FIG. 21.

(93) The state of the clearance or gaps between the driving interfaces of the impact wrench drive system components is not currently controlled and is nearly random. However, the effect of the clearances on the optimization of the system is important. When there is time for full transfer of energy to take place, the optimization actually simplifies greatly especially if that time can be assured. From the spring-mass oscillator model, it has been established that the anvil stiffness has a distinct effect on the timing of the energy. This is a powerful parameter to use in order to improve how quickly and completely energy is transferred in the case of very low or zero hex clearance. FIGS. 23 and 24 depict plots of the torque applied to the anvil (square-marked line) and the hex (star-marked line) to better show the timing of those peaks. The plot particularly depicted in FIG. 23 is a simulation with a small clearance in the hex and shows that the anvil is still deflected, and harboring energy, when the hex ($k_{sub.1}$ —star-marked line) peaks. The deflection of the hex “interrupts” the deflection of the anvil and reaches a peak before the anvil has released all its energy and has a predictable (but difficult to measure) interaction with it. The plot depicted in FIG. 24 shows the same hex clearance and an anvil that is about 4 times stiffer. The anvil (square-marked line) deflects to about 7000 in-lb and completely unloads prior to the hex ($k_{sub.1}$ —star-marked line) peak that occurs at about $2.00E-4$ seconds. The return of the anvil to the undeflected state indicates a separation or disengagement of the hammer from the anvil and an assurance that the anvil is not harboring significant potential energy during that disengagement. The model suggests that there is a minimum anvil stiffness required to achieve disengagement, and therefore complete energy transfer, for each state of hex clearance for any system. The incentive for using the minimum anvil stiffness is the reduced anvil torque (peak of the square-marked line) that the anvil has to be designed to durably withstand.

(94) A plot of simulated torque output as the clearance in the hex, called “hex gap” increases is depicted in FIG. 25. The plot shows that output of a dynamically tuned high stiffness spline drive anvil (solid line) is much less sensitive to the hex gap than the tuned low stiffness anvil (black dashed line). While anvils are both “tuned” to some degree, this plot assumes that the anvil stiffness was not part of the tuning process. Since the hex gap is very random, any given impact event will have output somewhere along these curves. The final performance will be some cumulative effect of all points achieved with a “ratcheting” effect in the case of a bolt tightening. The “ratcheting” effect occurs when the higher energy impacts are more effective even though they represent the minority of impacts. Obviously the flatter this curve is, the less scattered the productivity of the impacts will be and overall the tightening will achieve a higher torque.

(95) The presence of clearance in the hex also effects the optimization of inertia. Whether the anvil stiffness is able to be increased or not, if separation between the anvil and the hammer is the predominant type of impact event, then the theories of collision and momentum will apply.

Recalling the energy accounting and timing discussion of harbored anvil energy at low anvil stiffness, it is desirable to ensure that the hammer does not harbor energy (kinetic, in this case) when the torque in the hex reaches its peak. Otherwise, the energy is considered late and fails to contribute to the work done on the nut. Therefore, for optimal performance, the velocity of the hammer must be zero when the peak of the hex is reached. Assuming that other conditions are favorable for a disengagement scenario, such as a stiff anvil and/or an adequate hex clearance state, the proper inertia of the anvil/socket combination is equal to the inertia of the hammer. Consider the diagrams depicted in FIGS. 26-28 that use a billiards analogy to demonstrate the momentum factors.

(96) As depicted in FIG. 26, a striped ball approaches the white ball that is stationary. The striped ball has a significantly larger mass than the white ball. After the striped ball hits the white ball, the state after impact is shown below. The white ball moves with a significantly higher velocity than that with which the striped ball approached due to its low mass. The striped ball does not completely come to a stop for the same reason. Momentum equations bear this out. Subscripts “s” and “w” in the following equations indicate striped and white ball color while “i” and “f” indicate initial and final conditions with respect to the time of impact.

$$(97) m_s v_{si} + m_w v_{wi} = m_s v_{sf} + m_w v_{wf} \quad m_s (v_{si} - v_{sf}) = m_w (v_{wf} - v_{wi}) \quad \frac{m_s}{m_w} = \frac{(v_{wf} - v_{wi})}{(v_{si} - v_{sf})}$$

(98) The ratio of the masses dictates the ratio of the changes in velocities. Additionally, the continuing forward velocity of the striped ball and the spring toward which the white ball heads makes it likely that there will be additional contact between the balls before the striped ball's velocity becomes negative and heads in the opposite direction from which it approached. This bouncing behavior is highly inefficient and undesirable operation.

(99) If the white ball is significantly smaller than the striped ball then the striped ball will have a continuing (positive) velocity in the original approach direction, as in FIG. 26. If the white ball is much larger than the striped ball, as in FIG. 27, then the striped ball will have a negative velocity and head the opposite way that it approached. Either way, the striped ball is harboring kinetic energy that is not reaching the spring on the wall.

(100) The only way to get complete momentum and energy transfer is when the striped ball has zero velocity after the impact. Knowing that $V_{sub.wf}$ and $V_{sub.sf}$ are both zero and that $V_{sub.si}$ is NOT zero, the only way for this to be true is for $m_{sub.s}$ and $m_{sub.w}$ to be equal therefore making $V_{sub.wf} = V_{sub.si}$.

(101) ##STR00001##

(102) Therefore, for cases where the state of hex clearance is adequately large or the anvil is relatively stiff, the optimal socket/anvil inertia is equal to the hammer inertia, as depicted in FIG. 28.

(103) Since the hex clearance state at any impact event is relatively random, there will be conditions that will vary between a zero gap condition and an adequate gap condition. Prediction of performance will then have an upper bound defined by the momentum-based model and a lower bound defined by the spring-mass oscillation model. Likewise, the optimal inertia and stiffness will lie somewhere between the optimums dictated by the two models, as depicted in FIG. 29. If one condition is or can be made more likely, then following one model over the other might be advisable. It is therefore advantageous to attempt to choose parameters that would make the system less sensitive to the gap so that the upper and lower bound are very close and the application of one model over the other isn't as important.

(104) Dynamically Tuned Impact Wrench Drive System Components

(105) An impact wrench having dynamically tuned drive components may be capable of generating higher torque outputs, without increasing the weight, size or cost of the tool. The tuned drive components are optimized for inertial performance and stiffness, and are capable of transmitting energy more effectively and efficiently than standard impact wrench and socket designs. As such, a dynamically tuned impact wrench may solve the problem of achieving both high impact torques

while operating a maximum motor operating points, and may also prevent erratic operation while being operated at low mechanism speeds. The tuned drive components permit successful performance in both modes of operation (max motor and low speed), while incorporating lighter weight componentry having smaller size requirements. Another advantage obtained from utilizing an impact wrench having dynamically tuned drive components is a substantially advanced combination of extreme impact power and untethered portability. For example, extreme impact power has been obtained before, but has always been limited to pneumatic powered applications, which require an air hose to be connected to the tool and thusly restricting tool mobility. Dynamic tuning of the drive components facilitates the integration of reduction gearing in the drive train and permits the motor to more effectively run at high speed. Moreover, a tuned wrench obtains the advantage of increased power to weight ratio, since standard parts can be reduced in size, while still maintaining qualities of high performance and durability.

(106) When dynamically tuned, the socket and anvil are still separate components but are connected by an extremely stiff connection, such as a spline. The stiffness of the connection between the two drive components provides the following advantages over present solutions: 1) it causes the connected components to substantially behave as a single component such that the inertia of the anvil can simply be added to the socket inertia when determining optimal inertia. This means all the inertia required for optimal performance does not have to exist on the socket itself, but can be “hidden” further back inside the tool out of the immediate region of the fastener; and 2) a limiting factor in the overall stiffness of the anvil-socket combination has typically been the square shaped connection between the socket and the anvil. Increasing the socket/anvil connection stiffness allows the overall stiffness to be increased. As mentioned previously, increasing this stiffness reduces the inertia required to reach optimal performance. Tuning methodology, implemented through execution of at least one of two primary models (the spring-mass oscillator model and the momentum model) has rendered optimized performance characteristics with bounded ideal cases allowing for introduction and comparison of empirical test data, thereby facilitating part design optimized for multi-varied tool operation differences, such as looseness or clearance gaps between coupled components, as well as optimal structure changes in view of the balance between inertia ratios and stiffness ratios. For example, dynamic tuning reveals that the total stiffness of the anvil-socket combination including the interface between the two is in the region of 4/3 of the stiffness of the hex on which the tool is being used. Otherwise the inertia ratio for optimal performance at the minimum weight is a prescribed value in relation to the stiffness ratio.

(107) As depicted in FIGS. 30A-30C, and embodiment of a tuned anvil **6022** is provided for maximum stiffness and anvil strength. Unlike the tuned anvil **5022**, instead of a “neckdown” region where the stiffness can be manipulated during design, this anvil embodiment has a supporting flange **6071** that serves to stiffen the jaws **6087** by supporting them on the downstream side. The jaws **6087** no longer cantilever from the central hub alone, but are connected to the flange **6071** thereby increasing the stiffness. The integration of the jaws **6087** with the flange **6071** also serves to increase the strength of the jaws **6087** and increase life expectancy of the component part. Dynamic tuning can still promote design changes with respect to the diameter D3 and length L3 of the socket engagement portion **6047**, depending, to some extent, on modelled input and, corresponding test data.

(108) The tuned anvil **6022** may be mated to a correspondingly tuned socket **6010**, as depicted in exploded view in part of FIG. 31. As shown the tuned socket **6010** includes a mating portion having exterior splines configured to mate with complimentary splines of the mating portion of anvil **6022**. The addition of the sufficiently stiff anvil/socket coupling alters the diagram of FIG. 6 (associated with the standard ball and cam mechanism) to a model representing tuned components, as shown further in FIG. 31. Notably, the hammer **1020** and hex fastener **1** may remain unchanged.

(109) Similar modeling alteration is depicted in FIG. 32, which shows a swinging weight or Maurer

type hammer **3020** operable with a tuned anvil **7022** and correspondingly tuned socket **7010** to optimally drive a hex fastener **1**. Again, as the hammer **3020** design can remain unchanged, while the impact wrench drive system components are tuned for optimal performance in view of the inertia ratio and stiffness ratio of the anvil/socket combination.

(110) As discussed, there are several advantages obtained from dynamically tuning the drive components of an impact wrench. For example, one such advantage pertains to desirous changes in the external dimensions of tuned components, as depicted in FIG. **33**. Shown are three cordless (battery powered) impact wrenches. All three wrenches utilizes a common ball and cam hammer mechanism. However, dynamic tuning of other drive components renders benefits in both performance and look. As depicted, the middle impact wrench **2012** is engaged with a dynamically tuned socket **2010**—a power socket—and therefore performs with much higher torque output. However, to obtain the higher torque output the socket **2010** has a greatly enlarged diameter $D_{sub.P}$ attributable to the added annular inertia that renders higher performance. The top impact wrench **6012** takes advantages of dynamic tuning of not only the socket **6010**, but also the anvil in combination with the socket. The result is two-fold: higher torque output and smaller tool footprint, because the length of the socket **6010** (in functional combination with the anvil **6022**, not shown) is reduced by a distance $L_{sub.R}$, while the diameter of the socket **6010** remains the same $D_{sub.S}$, as a common socket **1010**. Hence, the advantages of dynamic tuning are not only evident in the performance of the tool, but are readily seen with regard to the reduced size of the tool.

(111) The engagement structure between the dynamically tuned socket and anvil has been primarily described and depicted as an involute spline with teeth that standard cutting tools in the industry can manufacture. A spline engagement is, therefore, desirable from the standpoint of both manufacturability and strength. However, there are alternatives that can also meet (or come close to meeting) the stiffness, inertial, and durability necessities pertinent to dynamically tuned impact wrench drive components. For example, FIG. **34** depicts several different engagement structures that may afford functionally operable stiffness when the corresponding structures of the tuned anvil and socket are connected. Features such as a triple square **47a**, a stub tooth spline **47b**, square teeth **47c**, arc teeth, **47d**, radial slots **47e**, tri-lobes, **47f**, hex indents **47g**, and keys and key ways, **47h**, may all provide sufficient structural functionality to comport with the optimal design characteristics revealed through dynamic tuning. Moreover, what has been depicted and described herein may apply to the socket side or anvil side. In other words, the anvil can include an external mating shape or an internal mating shape. An internal shape offers some advantages, because external structure can be utilized to maximize the amount of inertia that can exist on the anvil. However, an external mating structure on the anvil, such as external spline **47i**, can be designed to meet the tuned stiffness requirements and may be as, or almost as, effective from the standpoint of part performance

(112) While this disclosure has been described in conjunction with the specific embodiments outlined above, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, the preferred embodiments of the present disclosure as set forth above are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the present disclosure, as required by the following claims. The claims provide the scope of the coverage of the present disclosure and should not be limited to the specific examples provided herein.

Claims

1. A power tool assembly comprising: a housing configured to house a motor; a drive system including: a hammer assembly, the hammer assembly including a hammer configured to be driven by the motor; an anvil having an anvil jaw, the anvil jaw being periodically engaged by the hammer as the hammer is being driven, the anvil further including: a socket engagement portion located

opposite the anvil jaw, the socket engagement portion having a cylindrical wall forming a generally cylindrical open cavity and including interior splines disposed along an inner surface of the cylindrical wall; and a socket having a generally cylindrical first end and a second end, the socket first end defining an anvil engagement portion configured to be removably fitted directly into the open cavity of the socket engagement portion, the anvil engagement portion including exterior splines configured to engage with the interior splines of the socket engagement portion, the socket second end configured to receive a fastener opposite the anvil engagement portion; wherein the interior splines and the exterior splines provide stiffness between the anvil and the socket, and a total inertia of the drive system is split between the inertia of the hammer, the inertia of the anvil, and the inertia of the socket, and wherein an inertia ratio is decreased as a stiffness ratio increases, where the inertia ratio is the ratio of the combined inertias of the anvil and the socket to the inertia of the hammer assembly, and where the stiffness ratio is the ratio of the combined stiffness of the anvil and the socket to the stiffness of the fastener.

2. The power tool assembly of claim 1, wherein the interior splines and the exterior splines are selected from a group consisting of an involute spline, a triple square, a stub tooth spline, square teeth, arc teeth, radial slots, tri-lobes, hex indents, and keys and keyways.
3. The power tool assembly of claim 1, wherein the inertia ratio is between about 1.0 and about 2.0.
4. The power tool assembly of claim 3, wherein the stiffness ratio is between about 0.25 and about 0.75.
5. The power tool assembly of claim 1, wherein the hammer is a ball-and-cam type hammer.
6. The power tool assembly of claim 5, wherein the inertia ratio is between about 0.75 and about 1.0.
7. The power tool assembly of claim 6, wherein the stiffness ratio is between about 1.05 and about 1.55.
8. The power tool assembly of claim 7, wherein the stiffness ratio is about 1.33.
9. The power tool assembly of claim 1, wherein the socket engagement portion is integral to the anvil.
10. A drive system of an impact wrench comprising: a hammer assembly, the hammer assembly having a hammer configured to be driven by the motor; an anvil having an anvil jaw, the anvil jaw being periodically engaged by the hammer as the hammer is being driven, the anvil further including: a socket engagement portion located opposite the anvil jaw, the socket engagement portion having a cylindrical wall forming a generally cylindrical open cavity and including interior splines disposed along an inner surface of the cylindrical wall; and a socket having a generally cylindrical first end and a second end, the socket first end having a first diameter and defining an anvil engagement portion configured to be removably fitted directly into the open cavity of the socket engagement portion, the anvil engagement portion having exterior splines configured to engage with the interior splines of the socket engagement portion, the socket second end having a second diameter configured to receive a fastener opposite the anvil engagement portion; wherein the interior splines and the exterior splines provide stiffness between the anvil and the socket, and the total inertia of the drive system is divided between the inertia of the hammer, the inertia of the anvil, and the inertia of the socket, and wherein an inertia ratio is decreased as a stiffness ratio increases, where the inertia ratio is the ratio of the combined inertias of the anvil and the socket to the inertia of the hammer assembly, and where the stiffness ratio is the ratio of the combined stiffness of the anvil and the socket to the stiffness of the fastener.
11. The drive system of claim 10, wherein the interior splines and the exterior splines are selected from a group consisting of an involute spline, a triple square, a stub tooth spline, square teeth, arc teeth, radial slots, tri-lobes, hex indents, and keys and keyways.
12. The drive system of claim 10, wherein the inertia ratio is between about 1.0 and about 2.0.
13. The drive system of claim 12, wherein the stiffness ratio is between about 0.25 and about 0.75.
14. The drive system of claim 10, wherein the hammer is a ball-and-cam type hammer.

15. The drive system of claim 14, wherein the inertia ratio is between about 0.75 and about 1.0.
 16. The drive system of claim 15, wherein the stiffness ratio is between about 1.05 and about 1.55.
 17. The drive system of claim 16, wherein the stiffness ratio is about 1.33.
 18. The drive system of claim 10, wherein the socket engagement portion is integral to the anvil.
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